

SYMBOLS

Symbol	Means
Σ	"add them all up"
N / n	how many values: N = population, n = sample
$\mu / \bar{x}$	mean (average): μ population, $\bar{x}$ sample
σ / s	standard deviation: σ population, s sample
σ^2 / s^2	variance = (SD) ²
M, Q_1, Q_2, Q_3	median; quartiles at 25% / 50% / 75% ($Q_2 = M$)
IQR, LF, UF	$Q_3 - Q_1$; lower / upper fence (outlier cutoffs)
z	z-score = # of SDs from the mean
RF	relative frequency = part ÷ total (decimal)
$p / \hat{p}$	proportion: population / sample

Greek letter or capital N → POPULATION (everyone). Plain letter or small n → SAMPLE (some).

Every number answer: formula → plug in → arithmetic → boxed answer + units. RF as decimal (0.3000). Round probabilities to 4 decimals.

CH 1 — DATA COLLECTION

Population / Sample / Parameter / Statistic (Q1)

- Population** = the **WHOLE** group you want to know about
- Sample** = the smaller group actually measured (look for the number)
- Individual** = **ONE** member
- Parameter** = number describing the population — usually unknown
- Statistic** = number computed from the sample — the number they give you

P-arameter → P-opulation S-tatistic → S-ample

Q: "A nutritionist studies whether a new diet reduces cholesterol. She selects 80 participants and measures cholesterol before and after. After 12 weeks the average drop is 15 points. Identify the population, sample, individuals, parameter, statistic."

A: Population = all people who could use the diet · Sample = the 80 participants · Individuals = each participant · Parameter = the true mean drop for everyone (μ , unknown) · Statistic = the 15-point average drop ($\bar{x} = 15$)

Process: 1 state the question → 2 collect data → 3 describe it → 4 inference (sample → population).

Variable types (Q2) — answer all 3

1. Qualitative or Quantitative?

- Qualitative** = words / categories / labels. Includes rankings (1st, 2nd) and ID-type numbers (ZIP).
- Quantitative** = numbers you can average.

2. Discrete or Continuous? (qualitative → write "Not applicable")

- Discrete** = COUNTED, whole numbers ("number of...")
- Continuous** = MEASURED, decimals possible (time, temp, weight, height, \$, age, %)

3. Level — ask in order:

- Meaningful order? **NO** → **Nominal** (car type, brand, yes/no)
- Equal gaps between values? **NO** → **Ordinal** (rankings, S/M/L, agree...disagree, letter grade, education level)
- Does 0 mean "none"? **NO** → **Interval** (temperature °F/°C, year)
- 0 = none? **YES** → **Ratio** (counts, \$, time, weight, height, age)

Qualitative → Nominal or Ordinal.
Quantitative → Ratio, except temperature / year = Interval.

Q: "For each variable state: qualitative or quantitative; if quantitative, discrete or continuous; level of measurement."

Variable	Q/Q	D/C	Level
Car transmission (Q2a)	Qual	N/A	Nominal
Number of pets (Q2b)	Quant	Discrete	Ratio
Temperature °F (Q2c)	Quant	Continuous	Interval
Ranking top 5 players (Q2d)	Qual	N/A	Ordinal
Education level; Likert; young/middle/old	Qual	N/A	Ordinal
Phone type; provider; yes/no; ID	Qual	N/A	Nominal
Income \$; age (yrs); hours; grams; grade %	Quant	Continuous	Ratio
# students; # vending machines	Quant	Discrete	Ratio

Observational vs Experiment (Q3)

- Explanatory variable** = the possible cause (input); **Response** = the outcome measured
- Observational** = only watch / survey / record / track
- Experiment** = researcher **ASSIGNS** the treatment (randomly assigned, given drug/placebo, control group)

Observational → association only → causation = NO.
Experiment → causation = YES.

- Lurking variable** = not measured, but may be the real cause
- Census** = measure everyone · **SRS** = every group of n equally likely: number everyone 1–N, generate n random numbers

Q: "Researchers collect survey responses from college students about daily screen time and GPA. Observational or experiment? Explanatory variable? Response variable? Can causation be claimed?"

A: Observational (only surveyed) · Explanatory = daily screen time · Response = GPA · Causation: No.

Q: "68 kindergartners were randomly assigned to xylophone, singing, or listening groups; IQ tested after 4 months." → **A:** Experiment (researcher assigned groups) → causation can be claimed.

Bias (Q4) — ask in order

- Who got asked?** wrong/partial group → **Sampling bias** (only HQ, only readers, mall, online poll)

2. **Did they answer?** many didn't → **Nonresponse bias** (only 20% replied, refused, hung up)

3. **Are answers true?** no → **Response bias** (safe answers, leading question, lying)

Q: "A company surveys employees about remote work. Identify the bias: (1) survey is sent only to employees at headquarters; (2) only 20% respond, mostly those unhappy with remote work; (3) some give 'safe' answers they think management wants."

A: (1) Sampling bias · (2) Nonresponse bias · (3) Response bias

Sampling error = normal sample-vs-population difference. **Nonsampling error** = the biases above.

CH 2 — TABLES & GRAPHS

Category tables (Q5)

RF = count ÷ total · Degrees = RF × 360 · % = RF × 100
count = RF × total · missing RF = 1 – others

Checks: counts add to total · RFs add to 1 · degrees add to 360.

Q: "200 households were asked their favorite weekend activity: Sports 60, Movies 50, Shopping 40, Reading 30, Other 20. Fill in the relative frequencies and degree measures. Which is most popular? What percent prefer Reading? True/False: more than half prefer Sports or Movies."

A: Sports $60/200 = 0.3000 \rightarrow 0.3000 \times 360 = 108^\circ$ · Movies $50/200 = 0.2500 \rightarrow 90^\circ$ · Shopping $40/200 = 0.2000 \rightarrow 72^\circ$ · Reading $30/200 = 0.1500 \rightarrow 54^\circ$ · Other $20/200 = 0.1000 \rightarrow 36^\circ$ · Most popular = Sports (biggest count) · Reading = $0.15 \rightarrow 15\%$ · Sports or Movies = $(60+50)/200 = 0.55 = 55\% > 50\% \rightarrow \text{True}$

- Bar graph** = one bar per category, gaps between. **Axis starts at 0, equal widths, 2-D.** Pareto = tallest → shortest.
- Pie** = slices, angle = $RF \times 360^\circ$; parts of a whole. **Bar** to compare categories.
- Comparing two groups of different sizes → **use RF, not counts.**

Classes & histograms (Q6)

- Lower / upper class limit** = first / last number of the class (60–69 → 60, 69)
- Class width** = start of next class – start of this class (50 – 40 = 10, NOT 49 – 40)
- Histogram** = bars for classes, touching. Most frequent class = biggest count (answer with the interval).
- How many in a range → add counts. Percent → counts ÷ total × 100.

Q: "Quiz scores of 40 students: 40–49 (4), 50–59 (6), 60–69 (10), 70–79 (12), 80–89 (8). (a) Lower and upper limits of the 3rd class? (b) Class width? (c) What percent scored 70 or higher? (d) How many scored below 60? (e) Which class has the highest frequency?"

A: (a) 3rd class = 60–69 → lower 60, upper 69 · (b) $50 - 40 = 10$ · (c) $12 + 8 = 20 \rightarrow 20/40 = 50\%$ · (d) $4 + 6 = 10$ · (e) 70–79 (frequency 12)

CH 2 (continued)

Shape (Q7) — look where the TAIL points

- **Bell-shaped** = hump in middle, symmetric: heights, weights, IQ
- **Uniform** = flat: fair die, random digits
- **Skewed RIGHT** = tail right, a few very BIG values: salaries, house prices, waiting times
- **Skewed LEFT** = tail left, a few very SMALL values: easy exam scores

Skewed right → mean > median · Skewed left → mean < median · Symmetric → mean ≈ median

- Q:** "Identify the likely shape: (a) salaries where a few executives earn millions; (b) heights of 10-year-old boys; (c) outcomes of rolling a fair die 200 times; (d) restaurant waiting times, most short but a few very long."
- A:** (a) Skewed right · (b) Bell-shaped · (c) Uniform · (d) Skewed right
- **Stem-and-leaf** = stem = all digits but last, leaf = last digit (56 → 5 | 6); list every stem, leaves in order, add key.
 - **Time-series** = time on bottom, value on side, dots connected.
 - **Misleading graph** = axis not starting at 0, unequal bar widths, 3-D → fix: start at 0, equal widths.

CH 3 — NUMBERS THAT DESCRIBE DATA

Running example: "The following are the one-way commute times (minutes) of 10 employees at a company: 18, 22, 25, 28, 30, 31, 33, 35, 36, 40." All 10 employees → treat as a POPULATION (N = 10). Questions: population mean, variance, SD, median, mode, shape; then a sample of the first 5; five-number summary, IQR, fences, outliers; Empirical Rule.

Mean (Q8a, Q9a)

Average: add all values, divide by how many.

$\mu = \Sigma x \div N$ (population) · $\bar{x} = \Sigma x \div n$ (sample)

- Q:** "Find the population mean." **A:** $18+22+25+28+30+31+33+35+36+40 = 298 \rightarrow \mu = 298/10 = 29.8$ min
- Q:** "Select the first 5 observations as a sample (18, 22, 25, 28, 30). Find the sample mean." **A:** $123/5 \rightarrow \bar{x} = 24.6$ min

Median (Q8c)

Middle value of SORTED data.

1. Sort.
2. Count n.
3. **n odd** → middle value. **n even** → average the two middle values.

- Q:** "Find the population median." **A:** n = 10 (even) → 5th & 6th values = 30, 31 → $M = (30+31)/2 = 30.5$ min
- Odd example: 76, 80, 84, **86**, 90, 92, 94 → M = 86

Mode (Q8d)

Value that appears most. Two tie → both. Nothing repeats → "no mode".

- Q:** "Find the population mode." **A:** nothing repeats → no mode. (7, 12, 12, 20, 20, 25 → modes 12 and 20.)

Shape from mean vs median (Q8e)

mean > median → skewed right · mean < median → skewed left · equal → symmetric

- Q:** "Based on mean vs. median, describe the shape." **A:** mean 29.8 < median 30.5 → skewed left (slightly)

Resistant = not thrown off by extreme values: median & IQR yes; mean, range, SD no. Symmetric → report mean & SD. Skewed → median & IQR.

Range

Range = max – min

40 – 18 = 22 min

Variance & Standard Deviation (Q8b population, Q9b sample)

SD = how far values typically are from the mean.
Variance = SD².

Population: $\sigma^2 = \Sigma(x - \mu)^2 \div N$ · $\sigma = \sqrt{\sigma^2}$
Sample: $s^2 = \Sigma(x - \bar{x})^2 \div (n - 1)$ · $s = \sqrt{s^2}$
"population"/all → ÷ N · "sample" → ÷ (n - 1)

1. Find the mean.
2. Column: value – mean.
3. Column: square it.
4. Add the squares.
5. ÷ N (population) or ÷ (n - 1) (sample) → variance.
6. $\sqrt{}$ → standard deviation.

x	x – μ	(x – μ) ²
18	–11.8	139.24
22	–7.8	60.84
25	–4.8	23.04
28	–1.8	3.24
30	0.2	0.04
31	1.2	1.44
33	3.2	10.24
35	5.2	27.04
36	6.2	38.44
40	10.2	104.04
add	0 ✓	407.60

- Q:** "Find the population variance and standard deviation." **A:** table above → $\sigma^2 = 407.60/10 = 40.76$ min² · $\sigma = \sqrt{40.76} = 6.38$ min
- Q:** "Sample = first 5 observations (18, 22, 25, 28, 30). Find the sample variance." **A:** $\bar{x} = 24.6$; differences –6.6, –2.6, 0.4, 3.4, 5.4; squares 43.56 + 6.76 + 0.16 + 11.56 + 29.16 = 91.20 → $s^2 = 91.20/(5-1) = 91.20/4 = 22.8$ min² (÷4, not ÷5) · $s = \sqrt{22.8} = 4.77$

Calculator (1-Var Stats): **σx** = population SD, **Sx** = sample SD. Variance = square the UNROUNDED SD. Still write the table.

Empirical Rule (Q11) — bell-shaped only

68% within 1 SD · 95% within 2 SD · 99.7% within 3 SD
pieces: 34% | 13.5% | 2.35% | 0.15% (each side) · above mean = 50%

1. Write μ and σ (round σ to 1 decimal if his numbers match that).
2. Landmarks: $\mu \pm 1\sigma$, $\mu \pm 2\sigma$, $\mu \pm 3\sigma$.
3. Find the question's numbers among the landmarks.
4. Add the pieces between them.

10.6	17.0	23.4	29.8	36.2	42.6	49.0
2.35%	13.5%	34%	34%	13.5%	2.35%	

- Q:** "Use the population mean and SD. (a) About 95% of commute times lie between what two values? (b) What percent fall between 23.4 and 36.2 minutes? (c) Between 36.2 and 42.6 minutes?"
- A:** $\mu = 29.8$, $\sigma \approx 6.4$ (his numbers match σ rounded to 6.4). (a) 95% = 2 SDs → $29.8 \pm 2(6.4) = 29.8 \pm 12.8 \rightarrow 17.0$ to 42.6 · (b) $23.4 = \mu - 1\sigma$, $36.2 = \mu + 1\sigma \rightarrow 68\%$ · (c) $36.2 = \mu + 1\sigma$, $42.6 = \mu + 2\sigma \rightarrow (95 - 68)/2 = 13.5\%$
- Quick: above $\mu+1\sigma = 16\%$ · above $\mu+2\sigma = 2.5\%$ · $\mu+1\sigma$ to $\mu+3\sigma = 15.85\%$

z-score & percentile

$z = (\text{value} - \text{mean}) \div \text{SD}$ (bigger z = relatively higher)

- Q:** "Mean ACT is 18 with SD 6; mean SAT is 1000 with SD 200. Student A scored 27 on the ACT, Student B scored 1200 on the SAT. Who scored relatively higher?" **A:** A: $z = (27 - 18)/6 = 1.50$ · B: $z = (1200 - 1000)/200 = 1.00 \rightarrow$ **Student A**

95th percentile = 95% of data at or below. Median = 50th percentile.

Quartiles & five-number summary (Q10a)

1. Sort.
2. Q_2 = median of everything.
3. **Q_1 = median of lower half · Q_3 = median of upper half · n odd → leave the median OUT of both halves.**
4. **Five-number summary = Min, Q_1 , M, Q_3 , Max**

- Q:** "Find the five-number summary (min, Q_1 , median, Q_3 , max)." **A:** Lower half 18, 22, **25**, 28, 30 → $Q_1 = 25$ · Upper half 31, 33, **35**, 36, 40 → $Q_3 = 35$ · M = 30.5 → **18, 25, 30.5, 35, 40**
- Odd n: 76, 80, 84 | 86 | 90, 92, 94 → $Q_1 = 80$, $Q_3 = 92$
- 32, 76, 80, 84 | 86, 90, 92, 94 → $Q_1 = 78$, M = 85, $Q_3 = 91$

IQR, fences, outliers (Q10b, c)

$IQR = Q_3 - Q_1$ · $LF = Q_1 - 1.5 \cdot IQR$ · $UF = Q_3 + 1.5 \cdot IQR$
outlier = below LF or above UF

- Q:** "Find the IQR. Find the lower and upper fences. Identify any outliers." **A:** $IQR = 35 - 25 = 10$ · $1.5 \times 10 = 15$ · $LF = 25 - 15 = 10$ · $UF = 35 + 15 = 50$ · smallest 18 > 10 and largest 40 < 50 → **no outliers**
- With outlier (32, 76, 80, 84, 86, 90, 92, 94): $Q_1 = 78$, $Q_3 = 91$, $IQR = 13$, $1.5 \times 13 = 19.5 \rightarrow LF = 58.5$, $UF = 110.5 \rightarrow 32$ is an outlier

Boxplot

1. Five-number summary + fences.
2. Box from Q_1 to Q_3 , line at median.
3. Whiskers to smallest / largest values INSIDE the fences.
4. Outliers = separate *.

18	25	30.5	35	40
- - - - -	=====	=====	- - - - -	
Min	Q_1	M	Q_3	Max

Read shape: median centered + equal whiskers → symmetric · median left + long right whisker → skewed right.

DON'T LOSE POINTS

- Sort before median / quartiles.
- "Sample" → ÷ (n - 1). "Population" → ÷ N.
- SD asked → take the $\sqrt{}$. Write units (variance = min²).
- σx = population, Sx = sample. Never square a rounded SD.
- Odd n → median stays OUT of the halves.
- Class width = 50 – 40 = 10, not 49 – 40.
- RF as decimal (0.3000). Degrees need °.
- Observational → NO causation.
- Skew = direction of the TAIL.
- "No mode" / "Not applicable" are real answers.
- Show every step.